

FLAT-PACK BENCH: Evaluating Spatio-Temporal Understanding in Large Vision-Language Models through Furniture Assembly

Supplementary Material

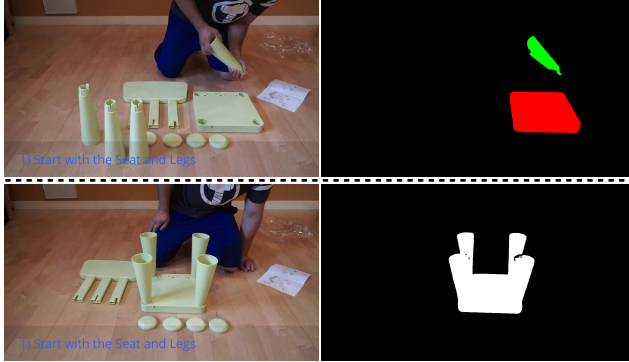


Figure S1. **Incomplete Segmentations in IMaW [14].** (Top) Segmentations are only provided for the parts that are about to be connected, i.e., the seat panel and the leg. Parts not being interacted with (the other legs and the backrest) are not annotated. (Bottom) Furthermore, instead of granular segmentations of each individual part, IMaW only has masks at the sub-assembly level.

A. Benchmark Details

A.1. Additional Example Questions

We provide some additional examples of the examples in our dataset in the supplementary HTML. Open the attached `index.html` and click on the link A.1 Additional Example Questions. You can toggle between *Key-frame* and *Trimmed* videos on the page. We also provide the responses of some of the models under the video and visual prompt settings that were used in Tab. 2.

A.2. Incomplete Segmentations in IMaW

In Sec. 3, we mentioned that IMaW [14] has incomplete segmentation annotations. Here, we elaborate this statement:

1. We found that segmentation annotations are only provided for parts that are in the process of being connected in a particular key frame. This limits the questions to these particular key frames, and even so, to only the parts annotated therein.
2. Furthermore, IMaW only includes information at the sub-assembly granularity, which precludes questions that one might want to ask about specific parts.

Figure S1 shows some examples of available segmentations in IMaW to illustrate these issues. Thus, we annotate our own segmentation maps as described in Sec. 3.

A.3. Manual Question Curation

Pitfalls of Auto-generated Questions. Figure S2 shows some examples of auto-generated questions, where the ques-

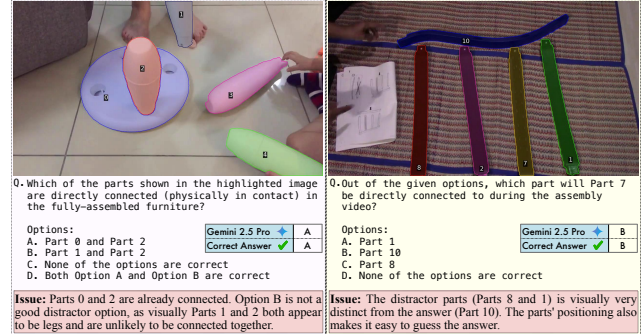


Figure S2. **Issues with Auto-generated Questions.** We show some examples to highlight the issue of auto-generated questions where the answer can be arrived at using the visual prompt alone and commonsense reasoning. Observe that we can arrive at the correct answer without requiring any temporal cues from the video.

tion can be solved using the visual prompt alone. In the first example (left), Part 0 is already connected to Part 2 in the visual prompt. Furthermore, Parts 1 and 2 are both clearly legs and unlikely to be connected together given the structure of Part 0. Overall, this frame is not suitable for constructing a MATE question. Similarly, in the second example, the distractor parts (Parts 1 and 8) appear to be quite clearly visually distinct from Part 10. The positioning of Parts 1 and 8 as if they are about to be inserted into Part 10 (not Part 7) also makes it easy to answer this question without any temporal context. Observe that even though we do not provide the videos for these questions, it is easy to arrive at the correct answer just with the visual prompt and some common-sense reasoning. Such examples prompted us to manually curate our questions.

Question Templates. As mentioned in Sec. 3, our annotators were given a list of question templates for creating the questions. Table S1 shows the full list of these question templates. Annotators must populate the parts highlighted in red with part IDs and provide options that they believe are challenging (parts with similar appearance, parts that are connected much later in the assembly compared to the state shown in the visual prompt, etc.) Most of the templates require up to 4 options, except for the second template in MATE, which is a binary question.

B. Evaluation Details

In this section, we provide additional experimental details for the evaluation in Sec. 4, such as details about video data processing, visual prompt construction, etc.

Task	Question Template
MATE	• Out of the given options, which part will <i>{query-part}</i> be directly connected to during the assembly video? • Are <i>{query-part1}</i> and <i>{query-part2}</i> connected in the fully-assembled furniture? • Which of the parts shown in the highlighted image are directly connected (physically in contact) in the fully-assembled furniture?
TLOC	• Which of the highlighted parts are connected (brought in direct physical contact) to <i>{query-part}</i> next starting from the assembly state shown in the image? • Which highlighted part is the last to connect (brought in direct physical contact) to <i>{query-part}</i> in the video? • Which highlighted pair of parts is connected first in the video? • Which highlighted pair of parts is connected last in the video? • Which labeled part was connected last to reach the current assembly state shown in the image? • Which labeled part will be connected next to the current assembly state shown in the image?
TORD	• Which of the following options lists the highlighted parts in the exact order that they become directly attached (physically in contact) to <i>{query-part}</i> in the video, from the first attachment event to the last? • Each option below lists a sequence of part connections. Which of these options gets the temporal ordering of the connections correct?
TRACK	• Identify the correct set of matches from Image A to Image B. • Identify which part in Image B matches Part <i>{query-part}</i> in Image A.

Table S1. **Question Templates.** We replace the **highlighted** part in the question template from scene to scene to construct our benchmark.

Video Data Processing. All videos are resized to a height of 480 px and width of 640 px. We dump all videos at 1 FPS before passing them as input to the model.

Video Subsampling. The videos in our benchmark are quite long (average duration of 6 minutes for key-frame videos at 1 FPS). Typically it is not possible to pass all the frames as input to models due to their limited context length. Hence, we uniformly subsample the frames before passing them as input. Subsampling frames uniformly from a fixed frame rate of 1 FPS is similar to the FPS-sampling strategy advocated by recent works [15, 44]. As for the number of frames to subsample, we followed the official guides/cookbooks released by the models wherever possible. Below we provide some details for the same:

- **Proprietary Models:** For Gemini [24], we provide the entire video as input. For trimmed videos which exceed the duration limit of 1 hour for Gemini, we dump the videos at the minimum FPS where the duration limit is satisfied. For GPT-5 [18], we provide 499 frames (498 in case of TRACK questions), as per the API token limit, which accomodates 500 frames.
- **Open Models:** For the Qwen family of models [1, 2] (including ArrowRL [30]), we sample a maximum of 768 frames, while maintaining a context length of 20480 tokens, as suggested by their official cookbook [1]. For all the other open models, except Video-LLaVA [12] (8 frames), we subsampled 32 frames, as per the setting used within their respective training and evaluation pipelines.

Visual Prompt Construction. As mentioned in Sec. 4, we use three different settings for constructing the visual prompt – *Mixed-Media*, *Collage*, and *Concat*. To see the full videos

in the supplementary HTML, open `index.html` and click on B. Evaluation Details.

The images that constitute the visual prompts are also resized to a resolution of 480×640 . For TRACK questions, the jumbled image prompt (Image B in Fig. 2) is provided as the *second* image in the input for *Mixed-Media*, as the middle image in each frame for *Collage*, and as the second frame of the video for *Concat*. These details are also conveyed to the model in the task instructions provided along with the input.

Task Instructions. In addition to the video, the visual prompts, and the question, the input contains task instructions that describe the input visual data to the model and specify the output format that the model should follow. Figures S3 to S5 show the task instructions that we use in our evaluations. When asking TRACK questions, we make adjustments to the input descriptions, informing the model that there are two visual prompts instead of one. Note that the task instructions are not a part of the benchmark. We welcome users of the benchmark to try out different phrasings of the instructions. For additional details and reproducibility, we will release the source code for our evaluations upon acceptance.

Final Prompt. The final prompt used in our evaluations has the following structure: [Video Frames] [Visual Prompt] [Jumbled Visual Prompt] [Task Instructions] [Question]. The [Jumbled Visual Prompt] is exclusive to TRACK questions.

Prompt for Human Participants. Figure S6 shows the instructions that human participants were provided before attempting our benchmark.

Task Instructions for Mixed-Media Prompts

You are a furniture-assembly expert. You are given:

1. A video of a furniture assembly in progress.
2. A single labeled frame from the same video, displaying numeric IDs on each visible furniture part. Call this a visual prompt.

For TRACK Questions

2. Two labeled frames from the same video, called Image A and Image B, each displaying numeric IDs on visible furniture parts. Note: The numeric IDs in Image A and Image B are not necessarily the same; the same ID may refer to different parts in each image.

3. A multiple-choice question (with its list of answer options) that refers to both the video and the image.

Some additional assumptions to keep in mind:

- Two furniture parts are "connected" if they are directly attached (physically in contact), like they would be in the final assembly.
- Parts simply touching each other physically, but not in their final assembly position, are not considered "connected".
- While deciding the order of "connection" events between two parts, consider the order in which they were connected last as their actual connection, i.e., if two parts are connected, then disconnected and then connected again, the last connection is the one that matters.
- In the course of the video, some parts may be attached physically to each other before they are fixed in their final assembly position by screwing, gluing, or otherwise securing them. In such cases, the parts are considered "connected" from the moment they are physically attached, not when they are fixed. Hence, when answering questions about the order of connections, consider the physical attachment as the connection event, even if the parts are not yet secured to each other.

Your responsibilities:

- Watch and interpret the assembly steps in the video.
- Examine the labeled image frame to understand part relationships.
- Read the question and all answer choices.
- Determine the correct option.
- Respond ****only**** with a JSON object containing a single key, "answer", whose value is the letter of your chosen option (e.g., "A", "B", "C").

****No additional text, explanations, or formatting -- just the JSON string.****

Now answer the following question:

Figure S3. **Task Instructions for the LVLMS.** We provide the models with task instructions that describe the input format, and additional information such as how we define `connection` events. When describing the inputs, we make adjustments for TRACK questions by mentioning that there are two visual prompts instead of one. The task instructions can be further broken down into 3 sections, describing the **Inputs**, **Assumptions**, and **Instructions** (colored accordingly above).

C. Additional Results

Human Performance. For the human performance results discussed in Sec. 4 We recorded a micro-average agreement of 80% indicating that our questions are clear and consistently understood. Task-wise, TORD attains the highest agreement (88%), followed by MATE (86%), and TRACK (77%), while TLOC attains the lowest (70%).

Zero-shot Results. In Sec. 4 we stated that we evaluated all models on every combination of video type (key-frame, trimmed) and visual prompt type (Mixed-Media, Concat, and Collage). Tab. 2 showed the results for the best settings

for each model. We provide the complete results in Tab. S3. An interactive version of the table is provided in the supplementary HTML, under C. Additional Results. If a setting is not shown, it was not evaluated due to cost constraints (e.g., GPT-5 [18] was assessed only on mixed-media prompts for key-frame videos) or because it does not support a prompt format (e.g., Mixed-Media prompts were not supported by some models, such as Video-LLaVA [12]).

Task Instructions for Collage Prompts

You are a furniture-assembly expert. You are given a video input where each frame consists of two images side-by-side:

- Right side: A frame from the furniture assembly video showing the assembly process.
- Left side: A single labeled frame from the video on the right side, fixed for the entire video, displaying numeric IDs on each visible furniture part.

The left-side labeled frame remains constant throughout the video.

You will be asked multiple-choice questions referring to the assembly video and the labeled parts shown on the left.

For TRACK Questions

You are a furniture-assembly expert. You are given a video input where each frame consists of three images side-by-side:

1. Right side: A frame from the furniture assembly video showing the assembly process.
2. Left side: A labeled frame from the video on the right side, fixed for the entire video, displaying bright numeric IDs on each visible furniture part. Call this Image A.
3. Center: Another labeled frame from the video on the right side, fixed for the entire video, also displaying bright numeric IDs on each visible furniture part. Call this Image B.

Both Image A and Image B remain constant throughout the video.

Note: The numeric IDs in Image A and Image B are not necessarily the same; the same ID may refer to different parts in each image.

You will be asked multiple-choice questions referring to the assembly video and the labeled parts shown on the left and central images.

• • •

Your responsibilities:

1. Use the right-side video frames to observe the assembly steps.
2. Use the fixed left-side labeled frame to identify and relate the furniture parts mentioned in the question.

For TRACK Questions

2. Use the fixed labeled frames on the left side (Image A) and in the center (Image B) to identify and relate the furniture parts mentioned in the question.

3. Carefully read the question and all answer choices.
4. Select the correct answer based on the video and labeled parts.
 - Respond ****only**** with a JSON object containing a single key, "answer", whose value is the letter of your chosen option (e.g., "A", "B", "C").

****Do not include any explanations or additional text -- reply with only the JSON string.****

Now answer the following question:

Figure S4. **Task Instructions for Collage Prompts.** *Assumptions* remain the same (denoted by . . .), but the *Inputs* and *Instructions* are modified to describe the prompt format. In this case, the prompt image(s) are attached to the left of the video frame. For TRACK questions, the jumbled visual prompt is the second (middle) image in each frame.

713 D. Analysis Details

714 D.1. Self-Explanations

715 We have provided some additional examples of self-
716 explanations, expanding upon the results discussed
717 in Sec. 5.4. For viewing them navigate to D.1
718 Self-Explanations in the supplementary HTML. We
719 also provide the full breakdown of error categories for the
720 generated rationales discussed in Sec. 5.4 in Fig. S7.

721 D.2. TVA Results

722 We discussed an agentic baseline TVA in Sec. 5.5. In this
723 section, we provide the experimental details for the frame-
724 work, along with additional qualitative examples.

D.2.1. Experimental Details

Contact-Reasoning Issues. For our contact-reasoning experiment, we use the task instructions from the image-only prompting experiments in App. D.4. We used two versions of the question template:

1. In this task, "connected" means the two parts are in direct physical contact, in the same way they will be when the furniture is fully assembled (not merely near each other or partially aligned). Based on the given image, are {query-part1} and {query-part2} connected?
2. Are {query-part1} and {query-part2} connected (physically in contact) in the shown image?

Table S2 shows the results. We observed poor perfor-

Task Instructions for Concat Prompts

You are a furniture-assembly expert. You are given a video input of the assembly process of a furniture item. The first frame of the video is a frame taken from the assembly process, highlighted with each visible furniture part, and displaying numeric IDs on each furniture part.

You will be asked multiple-choice questions referring to the assembly video and the labeled parts shown on the left.

For TRACK Questions

You are a furniture-assembly expert. You are given a video input of the assembly process of a furniture item. The first two frames of the video are frames taken from the assembly process, highlighted with each visible furniture part, and displaying numeric IDs on each furniture part. We will refer to them as Image A (the first frame) and Image B (the second frame).

Note: The numeric IDs in Image A and Image B are not necessarily the same; the same ID may refer to different parts in each image.

You will be asked multiple-choice questions referring to the assembly video and the labeled parts shown in the first frame.

...

Your responsibilities:

1. Use the video frames after the first frame to observe the assembly steps.
2. Use the first labeled frame to identify and relate the furniture parts mentioned in the question.

For TRACK Questions

1. Use the video frames after the first two frames to observe the assembly steps.
2. Use the first two labeled frames, i.e., Image A and Image B, to identify and relate the furniture parts mentioned in the question.

3. Carefully read the question and all answer choices.
4. Select the correct answer based on the video and labeled parts.
 - Respond ****only**** with a JSON object containing a single key, "answer", whose value is the letter of your chosen option (e.g., "A", "B", "C").

****Do not include any explanations or additional text -- reply with only the JSON string.****

Now answer the following question:

Figure S5. **Task Instructions for Concat Prompts.** *Assumptions* remain the same (denoted by . . .), but the *Inputs* and *Instructions* are modified to describe the prompt format. In this case, the prompt image(s) are concatenated as the initial frames of the video. For TRACK questions, the jumbled visual prompt is the second frame in the video.

1. Each question consists of:
 - a. A furniture assembly video.
 - b. 1-2 visual prompts or frames extracted from the video, with certain parts shaded and labeled.
 - c. An MCQ question with at most 4 options.
 2. Examine the labeled frame(s) and understand the relationships between the highlighted parts using the video.
 3. Read the question and then determine the correct option.

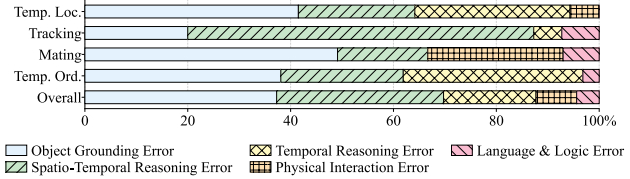


Figure S7. **Error Types for Rationales.** We provide the full quantitative distribution of error types for the rationales generated by Gemini 2.5 Pro, as discussed in Sec. 5.4.

Figure S6. **Human Evaluation Instructions.** Human participants were provided the above instructions along with the assembly video, visual prompt, and the question text.

mance across all settings, indicating that LVLMs struggle to understand even simpler concepts like physical contact.

Tracking Issues. We used SAM2 [19] (sam2.1_hiera_l) for our tracking experiments. We used trimmed videos for

this experiment. SAM2 cannot handle very long videos, so we implemented a chunk-wise propagation algorithm. First, we split the video into 256-frame clips with a single frame overlap between adjacent clips. Then, we first ran SAM2 on the clip containing the prompt frame. After this, we propagate the masks forward in the video by using the masks predicted in the last frame of the initial clip to prompt the first frame in the next clip. Similarly, we propagate the masks backward from the initial clip by using the masks predicted

Table S2. **Contact-Reasoning Results.** We show the performance of Qwen2.5-VL (32B & 72B) across two question templates. The overall performance is quite poor across all settings.

Model	Template	Micro Avg.	Yes	No
Qwen-2.5VL-32B	1	63.53	54.13	72.93
Qwen-2.5VL-32B	2	61.00	41.07	80.93
Qwen-2.5VL-72B	1	63.93	43.87	78.93
Qwen-2.5VL-72B	2	58.87	33.20	84.53

in the first frame of the initial clip to prompt the last frame in the preceding clip. We continue this process until the entire video has been covered. We subsampled frames by a factor of 4 to maintain a trade-off between compute constraints and temporal continuity.

D.2.2. Qualitative Examples

We provide additional qualitative examples for TVA in the supplementary HTML. Click on the attached `index.html` file and click on D.2.2 TVA Qualitative Results. You can view the question that we presented to the agent, along with the associated assembly video and the visual prompt constructed from the part masks. We also provide a walkthrough video that traces the execution of the code generated by the *Code LLM* and highlights the issues with the agents’ tools that limit its performance.

D.3. Linguistic Prompt Engineering

This section provides additional details about the linguistic prompt engineering experiments discussed in Sec. 5.1. Initially, we added “*Please explain this answer step-by-step*” to the task instruction. However, we found that the model was just paraphrasing its selected option. Hence, we expanded this instruction to explicitly push the model to explain its reasoning. The results in Sec. 5.1, and the qualitative results discussed in App. D.1 are from this modified prompt. We also amended the *Instructions* portion of the task instruction to add an `explanation` key to the output JSON. See Fig. S8 for the complete task instructions. In SC-CoT [26], for temperature sampling, we set the temperature to 0.7, and `top_k` and `top_p` set to 1 and 40, respectively [32].

D.4. Image-only Prompts

In this section, we discuss the experimental details for the image-only prompts discussed in Sec. 5.3. Figure S9 shows the task instruction used for our image-only experiments. The remaining settings were the same as our main zero-shot evaluation in Sec. 4.

Task Instructions for CoT Prompting

You are a furniture-assembly expert. You are given:

1. A video of a furniture assembly in progress.
2. A single labeled frame from the same video, displaying numeric IDs on each visible furniture part. Call this a visual prompt.

For TRACK Questions

2. Two labeled frames from the same video, called Image A and Image B, each displaying numeric IDs on visible furniture parts. Note: The numeric IDs in Image A and Image B are not necessarily the same; the same ID may refer to different parts in each image.

3. A multiple-choice question (with its list of answer options) that refers to both the video and the image.

Some additional assumptions to keep in mind:

- Two furniture parts are "connected" if they are directly attached (physically in contact), like they would be in the final assembly.
- Parts simply touching each other physically, but not in their final assembly position, are not considered "connected".
- While deciding the order of "connection" events between two parts, consider the order in which they were connected last as their actual connection, i.e., if two parts are connected, then disconnected and then connected again, the last connection is the one that matters.
- In the course of the video, some parts may be attached physically to each other before they are fixed in their final assembly position by screwing, gluing, or otherwise securing them. In such cases, the parts are considered "connected" from the moment they are physically attached, not when they are fixed. Hence, when answering questions about the order of connections, consider the physical attachment as the connection event, even if the parts are not yet secured to each other.

Your responsibilities:

- Watch and interpret the assembly steps in the video.
- Examine the labeled image frame to understand part relationships.
- Read the question and all answer choices.
- Determine the correct option.
- Please explain your answer step-by-step. This explanation should contain your reasoning process, including how you analyzed the video and image to arrive at your conclusion. Do not simply restate the question or the answer choices.
- Respond ****only**** with a JSON object containing two keys -
 - A) "answer", whose value is the letter of your chosen option (for example "A", "B", "C", etc.).
 - B) "explanation", whose value is a step-by-step explanation of your answer. This must not be a restatement of the question or the answer choices, but rather a detailed reasoning process that leads to your answer. For example, describe how you analyzed the video and image to arrive at your conclusion or why certain options were eliminated.

****Do not include any explanations or additional text - reply with only the JSON string.****

Now answer the following question:

Figure S8. **Task Instructions for CoT.** We ask the model to explain its response, explicitly asking it not to paraphrase the correct option.

Task Instructions for Image-only Experiments

You are a furniture-assembly expert. You are given:

1. A still image showing a step in the assembly process of a furniture item, with the furniture parts shaded and labeled, displaying numeric IDs on each visible furniture part. Call this a visual prompt.

For TRACK Questions

2. Two still images, Image A and Image B showing different steps in the assembly process of a furniture item, with the furniture parts shaded and labeled, displaying numeric IDs on each visible furniture part.

2. A multiple-choice question (with its list of answer options) that refers to the visual prompt.

Some additional assumptions to keep in mind:

- Two furniture parts are "connected" if they are directly attached (physically in contact), like they would be in the final assembly.
- Parts simply touching each other physically, but not in their final assembly position, are not considered "connected".

Your responsibilities:

- Observe and interpret the assembly step shown in the image.

For TRACK Questions

- Observe and interpret the assembly steps shown in the images.
- Examine the labeled image frames to understand part relationships.
- Read the question and all answer choices.
- Determine the correct option.
- Respond ****only**** with a JSON object containing a single key, `"answer"`, whose value is the letter of your chosen option (e.g., `"A"`, `"B"`, `"C"`).

****Do not include any explanations or additional text - reply with only the JSON string.****

Now answer the following question:

Figure S9. **Image-only Task Instructions.** For image-only questions, the **Inputs** only consist of 1-2 prompt images. We remove any mention of the video from the **Assumptions** and the **Instructions**.

Table S3. **Full Zero-shot Results on FLAT-PACK BENCH.** In this table, we show the performance of all the models that we evaluated on each setting of video type and visual prompt type.

Model	Prompt Type	Video Type	Micro Avg.	TORD	TLOC	TRACK	MATE
LLaVA-Video-7B	Collage	Key-frame	21.26	22.58	25.24	11.67	42.53
LLaVA-Video-7B	Collage	Trimmed	20.76	23.23	21.36	12.06	41.38
LLaVA-Video-7B	Concat	Key-frame	21.26	23.23	24.27	12.84	39.08
LLaVA-Video-7B	Concat	Trimmed	21.26	23.23	24.27	12.84	39.08
LLaVA-Video-7B	Mixed-media	Key-frame	24.42	23.87	32.04	12.06	52.87
LLaVA-Video-7B	Mixed-media	Trimmed	24.25	23.87	32.04	11.67	52.87
InternVL3-14B	Collage	Key-frame	37.21	41.29	22.33	36.96	48.28
InternVL3-14B	Collage	Trimmed	36.54	38.71	26.21	35.80	47.13
InternVL3-14B	Concat	Key-frame	34.05	38.06	29.13	26.85	54.02
InternVL3-14B	Concat	Trimmed	32.72	37.42	26.21	26.07	51.72
InternVL3-14B	Mixed-media	Key-frame	34.22	38.06	36.89	30.35	35.63
InternVL3-14B	Mixed-media	Trimmed	35.38	38.71	39.81	30.35	39.08
InternVL3-38B	Collage	Key-frame	34.05	36.13	35.92	25.29	54.02
InternVL3-38B	Collage	Trimmed	33.39	35.48	33.01	24.90	55.17
InternVL3-38B	Concat	Key-frame	28.57	38.06	29.13	15.18	50.57
InternVL3-38B	Concat	Trimmed	28.74	35.48	29.13	16.73	51.72
InternVL3-38B	Mixed-media	Key-frame	33.22	45.81	47.57	16.34	43.68
InternVL3-38B	Mixed-media	Trimmed	31.73	45.81	40.78	15.56	43.68
InternVL3-78B	Collage	Key-frame	36.54	30.32	34.95	41.25	35.63
InternVL3-78B	Collage	Trimmed	35.88	30.97	34.95	39.30	35.63
InternVL3-78B	Concat	Key-frame	38.21	34.19	32.04	42.80	39.08
InternVL3-78B	Concat	Trimmed	38.21	30.97	33.98	42.80	42.53
InternVL3-78B	Mixed-media	Key-frame	36.05	33.55	46.60	34.63	32.18
InternVL3-78B	Mixed-media	Trimmed	36.21	36.13	45.63	35.41	27.59
Qwen2.5-VL-32B	Collage	Key-frame	33.89	32.90	29.13	29.57	54.02
Qwen2.5-VL-32B	Collage	Trimmed	30.56	33.55	24.27	23.35	54.02
Qwen2.5-VL-32B	Concat	Key-frame	27.57	29.68	33.01	17.12	48.28
Qwen2.5-VL-32B	Concat	Trimmed	25.58	32.26	29.13	11.67	50.57
Qwen2.5-VL-32B	Mixed-media	Key-frame	31.06	37.42	33.01	21.79	44.83
Qwen2.5-VL-32B	Mixed-media	Trimmed	34.05	37.42	42.72	25.29	43.68
Qwen2.5-VL-72B	Collage	Key-frame	37.54	32.26	25.24	41.25	50.57
Qwen2.5-VL-72B	Collage	Trimmed	34.22	36.77	26.21	31.13	48.28
Qwen2.5-VL-72B	Concat	Key-frame	34.72	33.55	28.16	31.52	54.02
Qwen2.5-VL-72B	Concat	Trimmed	30.23	36.77	30.10	19.07	51.72
Qwen2.5-VL-72B	Mixed-media	Key-frame	43.52	38.71	37.86	50.19	39.08
Qwen2.5-VL-72B	Mixed-media	Trimmed	39.87	38.71	34.95	43.58	36.78
Qwen2.5-VL-7B	Collage	Key-frame	25.42	30.97	26.21	16.73	40.23
Qwen2.5-VL-7B	Collage	Trimmed	24.75	30.32	25.24	16.73	37.93
Qwen2.5-VL-7B	Concat	Key-frame	29.57	34.84	21.36	25.29	42.53
Qwen2.5-VL-7B	Concat	Trimmed	25.91	30.97	28.16	17.90	37.93
Qwen2.5-VL-7B	Mixed-media	Key-frame	30.56	27.74	23.30	31.52	41.38
Qwen2.5-VL-7B	Mixed-media	Trimmed	29.24	31.61	27.18	24.51	41.38

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Model	Prompt Type	Video Type	Micro Avg.	TORD	TLOC	TRACK	MATE
Qwen3-VL-235B-A22B	Collage	Key-frame	36.71	38.06	33.01	34.24	45.98
Qwen3-VL-235B-A22B	Collage	Trimmed	35.88	37.42	25.24	33.85	51.72
Qwen3-VL-235B-A22B	Concat	Key-frame	35.05	39.35	37.86	28.79	42.53
Qwen3-VL-235B-A22B	Concat	Trimmed	34.22	36.13	32.04	28.02	51.72
Qwen3-VL-235B-A22B	Mixed-media	Key-frame	33.72	38.71	41.75	27.24	34.48
Qwen3-VL-235B-A22B	Mixed-media	Trimmed	31.23	36.13	39.81	23.74	34.48
Qwen3-VL-30B-A3B	Collage	Key-frame	34.55	31.61	28.16	36.58	41.38
Qwen3-VL-30B-A3B	Collage	Trimmed	35.71	32.26	24.27	36.96	51.72
Qwen3-VL-30B-A3B	Concat	Key-frame	34.88	29.03	25.24	40.08	41.38
Qwen3-VL-30B-A3B	Concat	Trimmed	36.54	34.19	22.33	40.08	47.13
Qwen3-VL-30B-A3B	Mixed-media	Key-frame	36.38	34.19	35.92	36.58	40.23
Qwen3-VL-30B-A3B	Mixed-media	Trimmed	35.71	34.19	31.07	38.91	34.48
Qwen3-VL-32B	Collage	Key-frame	35.55	34.84	33.01	34.63	42.53
Qwen3-VL-32B	Collage	Trimmed	34.55	41.29	27.18	31.13	41.38
Qwen3-VL-32B	Concat	Key-frame	33.06	32.90	30.10	29.96	45.98
Qwen3-VL-32B	Concat	Trimmed	32.06	36.13	28.16	26.85	44.83
Qwen3-VL-32B	Mixed-media	Key-frame	36.54	37.42	46.60	29.96	42.53
Qwen3-VL-32B	Mixed-media	Trimmed	35.71	37.42	46.60	28.40	41.38
Qwen3-VL-32B-Think	Collage	Key-frame	34.55	38.71	26.21	31.52	45.98
Qwen3-VL-32B-Think	Collage	Trimmed	39.87	37.42	23.30	45.14	48.28
Qwen3-VL-32B-Think	Concat	Key-frame	33.72	40.65	32.04	26.85	43.68
Qwen3-VL-32B-Think	Concat	Trimmed	34.72	33.55	22.33	35.02	50.57
Qwen3-VL-32B-Think	Mixed-media	Key-frame	38.87	42.58	36.89	36.96	40.23
Qwen3-VL-32B-Think	Mixed-media	Trimmed	37.54	39.35	37.86	35.41	40.23
Qwen3-VL-4B	Collage	Key-frame	33.39	30.97	32.04	27.63	56.32
Qwen3-VL-4B	Collage	Trimmed	35.22	30.97	33.01	32.68	52.87
Qwen3-VL-4B	Concat	Key-frame	33.39	31.61	26.21	30.74	52.87
Qwen3-VL-4B	Concat	Trimmed	35.55	30.32	35.92	30.35	59.77
Qwen3-VL-4B	Mixed-media	Key-frame	34.22	30.97	30.10	33.46	47.13
Qwen3-VL-4B	Mixed-media	Trimmed	34.55	29.68	32.04	33.85	48.28
Qwen3-VL-4B-Think	Collage	Key-frame	27.57	30.32	26.21	19.07	49.43
Qwen3-VL-4B-Think	Collage	Trimmed	29.73	32.26	25.24	24.12	47.13
Qwen3-VL-4B-Think	Concat	Key-frame	32.72	34.19	27.18	28.02	50.57
Qwen3-VL-4B-Think	Concat	Trimmed	38.21	34.84	28.16	39.69	51.72
Qwen3-VL-4B-Think	Mixed-media	Key-frame	27.08	29.03	28.16	21.40	39.08
Qwen3-VL-4B-Think	Mixed-media	Trimmed	29.73	27.74	35.92	24.51	41.38
Qwen3-VL-8B	Collage	Key-frame	27.91	32.26	32.04	21.01	35.63
Qwen3-VL-8B	Collage	Trimmed	24.92	30.32	26.21	16.34	39.08
Qwen3-VL-8B	Concat	Key-frame	27.41	30.32	31.07	20.23	39.08
Qwen3-VL-8B	Concat	Trimmed	28.41	30.97	24.27	24.51	40.23
Qwen3-VL-8B	Mixed-media	Key-frame	33.06	33.55	32.04	33.07	33.33
Qwen3-VL-8B	Mixed-media	Trimmed	31.40	32.26	33.98	31.52	26.44
Qwen3-VL-8B-Think	Collage	Key-frame	28.57	31.61	29.13	20.62	45.98
Qwen3-VL-8B-Think	Collage	Trimmed	29.40	35.48	29.13	19.84	47.13
Qwen3-VL-8B-Think	Concat	Key-frame	25.91	34.19	31.07	12.45	44.83

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Model	Prompt Type	Video Type	Micro Avg.	TORD	TLOC	TRACK	MATE
Qwen3-VL-8B-Think	Concat	Trimmed	28.74	36.77	23.30	17.12	55.17
Qwen3-VL-8B-Think	Mixed-media	Key-frame	26.74	33.55	29.13	19.46	33.33
Qwen3-VL-8B-Think	Mixed-media	Trimmed	31.56	34.19	36.89	23.35	44.83
PerceptionLM-1B	Collage	Key-frame	26.91	28.39	23.30	24.51	35.63
PerceptionLM-1B	Collage	Trimmed	26.74	27.10	23.30	25.29	34.48
PerceptionLM-1B	Concat	Key-frame	28.07	23.87	31.07	26.85	35.63
PerceptionLM-1B	Concat	Trimmed	28.41	25.16	29.13	28.40	33.33
PerceptionLM-3B	Collage	Key-frame	29.57	28.39	33.01	26.07	37.93
PerceptionLM-3B	Collage	Trimmed	30.90	28.39	32.04	28.79	40.23
PerceptionLM-3B	Concat	Key-frame	29.24	27.74	28.16	29.18	33.33
PerceptionLM-3B	Concat	Trimmed	30.23	29.68	34.95	27.24	34.48
PerceptionLM-8B	Collage	Key-frame	34.39	25.81	25.24	43.19	34.48
PerceptionLM-8B	Collage	Trimmed	35.05	26.45	27.18	43.97	33.33
PerceptionLM-8B	Concat	Key-frame	31.56	25.16	23.30	38.91	31.03
PerceptionLM-8B	Concat	Trimmed	30.90	24.52	24.27	36.96	32.18
Gemini 2.5 Flash	Collage	Key-frame	24.09	27.10	36.89	12.06	39.08
Gemini 2.5 Flash	Collage	Trimmed	19.27	21.29	28.16	11.67	27.59
Gemini 2.5 Flash	Concat	Key-frame	28.90	36.13	44.66	12.84	44.83
Gemini 2.5 Flash	Concat	Trimmed	22.09	26.45	33.01	8.95	40.23
Gemini 2.5 Flash	Mixed-media	Key-frame	31.23	32.90	41.75	22.96	40.23
Gemini 2.5 Flash	Mixed-media	Trimmed	26.58	30.97	33.98	18.68	33.33
Gemini 2.5 Pro	Collage	Key-frame	28.12	17.39	35.71	31.25	36.36
Gemini 2.5 Pro	Concat	Key-frame	33.85	39.13	57.14	17.65	18.18
Gemini 2.5 Pro	Mixed-media	Key-frame	34.72	43.23	44.66	23.74	40.23
GPT-5	Mixed-media	Key-frame	38.04	40.65	53.40	26.46	49.43
LLaVA-NeXT-Vid-34B	Collage	Key-frame	27.41	29.68	25.24	28.02	24.14
LLaVA-NeXT-Vid-34B	Collage	Trimmed	26.41	30.97	23.30	26.07	22.99
LLaVA-NeXT-Vid-34B	Concat	Key-frame	26.91	32.26	20.39	27.24	24.14
LLaVA-NeXT-Vid-34B	Concat	Trimmed	27.24	33.55	19.42	27.24	25.29
LLaVA-NeXT-Vid-34B	Mixed-media	Key-frame	28.57	26.45	25.24	31.91	26.44
LLaVA-NeXT-Vid-34B	Mixed-media	Trimmed	29.90	29.03	24.27	32.30	31.03
LLaVA-NeXT-Vid-7B	Collage	Key-frame	23.92	29.03	27.18	16.73	32.18
LLaVA-NeXT-Vid-7B	Collage	Trimmed	24.92	30.97	25.24	17.51	35.63
LLaVA-NeXT-Vid-7B	Concat	Key-frame	24.42	29.68	31.07	12.06	43.68
LLaVA-NeXT-Vid-7B	Concat	Trimmed	24.09	29.03	30.10	12.06	43.68
LLaVA-NeXT-Vid-7B	Mixed-media	Key-frame	24.58	25.81	26.21	17.51	41.38
LLaVA-NeXT-Vid-7B	Mixed-media	Trimmed	24.42	27.10	26.21	15.18	44.83
LLaVA-OV-72B	Collage	Key-frame	37.87	33.55	31.07	36.58	57.47
LLaVA-OV-72B	Collage	Trimmed	38.21	34.19	30.10	37.35	57.47
LLaVA-OV-72B	Concat	Key-frame	37.04	35.48	27.18	36.19	54.02
LLaVA-OV-72B	Concat	Trimmed	36.71	36.13	25.24	35.41	55.17
LLaVA-OV-7B	Collage	Key-frame	29.07	23.87	24.27	30.74	39.08
LLaVA-OV-7B	Collage	Trimmed	28.24	23.87	19.42	29.57	42.53
LLaVA-OV-7B	Concat	Key-frame	32.89	25.81	29.13	35.02	43.68
LLaVA-OV-7B	Concat	Trimmed	32.72	25.16	30.10	35.02	42.53

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Model	Prompt Type	Video Type	Micro Avg.	TORD	TLOC	TRACK	MATE
LLaVA-OV-7B	Mixed-media	Key-frame	31.40	27.74	29.13	33.85	33.33
LLaVA-OV-7B	Mixed-media	Trimmed	31.40	28.39	26.21	34.24	34.48
LLaVA-Video-72B	Collage	Key-frame	37.21	34.84	29.13	35.41	56.32
LLaVA-Video-72B	Collage	Trimmed	34.55	34.84	22.33	33.85	50.57
LLaVA-Video-72B	Concat	Key-frame	32.23	34.19	29.13	24.51	55.17
LLaVA-Video-72B	Concat	Trimmed	32.39	35.48	33.01	22.96	54.02
LLaVA-Video-72B	Mixed-media	Key-frame	34.39	32.90	28.16	30.74	55.17
LLaVA-Video-72B	Mixed-media	Trimmed	34.05	36.77	28.16	28.40	52.87
LLaVA-Video-7B	Collage	Key-frame	27.74	21.94	25.24	23.74	52.87
LLaVA-Video-7B	Collage	Trimmed	28.57	25.81	26.21	25.29	45.98
LLaVA-Video-7B	Concat	Key-frame	32.56	27.74	26.21	32.30	49.43
LLaVA-Video-7B	Concat	Trimmed	33.39	26.45	25.24	34.24	52.87
LLaVA-Video-7B	Mixed-media	Key-frame	29.73	27.10	28.16	24.90	50.57
LLaVA-Video-7B	Mixed-media	Trimmed	28.74	25.16	25.24	24.12	52.87
ArrowRL-7B	Collage	Key-frame	28.74	34.84	30.10	20.23	41.38
ArrowRL-7B	Collage	Trimmed	23.75	26.45	23.30	17.90	36.78
ArrowRL-7B	Concat	Key-frame	29.73	34.84	29.13	22.96	41.38
ArrowRL-7B	Concat	Trimmed	27.08	28.39	29.13	21.79	37.93
ArrowRL-7B	Mixed-media	Key-frame	28.41	31.61	24.27	26.07	34.48
ArrowRL-7B	Mixed-media	Trimmed	29.73	29.03	31.07	26.85	37.93
Gemini 2.5 Pro + GenS	Collage	Key-frame	26.41	34.84	34.95	13.23	40.23
Gemini 2.5 Pro + GenS	Concat	Key-frame	31.40	40.00	41.75	19.46	39.08

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References

- [1] Qwen3-vl github repository. <https://github.com/QwenLM/Qwen3-VL>, 2025. Accessed: 2025-10-20. 4, 5, 2
- [2] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. 1, 4, 5, 6, 2
- [3] Fabian Caba Heilbron, Victor Escorcia, Bernard Ghanem, and Juan Carlos Niebles. Activitynet: A large-scale video benchmark for human activity understanding. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 961–970, 2015. 1, 2
- [4] Jang Hyun Cho, Andrea Madotto, Effrosyni Mavroudi, Triantafyllos Afouras, Tushar Nagarajan, Muhammad Maaz, Yale Song, Tengyu Ma, Shuming Hu, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Suyog Jain, Miguel Martin, Huiyu Wang, Nikhila Ravi, Shashank Jain, Temmy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, and Christoph Feichtenhofer. Perceptionlm: Open-access data and models for detailed visual understanding. *arXiv:2504.13180*, 2025. 1, 2, 4, 5
- [5] Google AI for Developers. Gemini thinking — gemini api documentation, 2025. Accessed: 2025-11-13. 7
- [6] Google DeepMind. Gemini 2.5 pro model card. <https://modelcards.withgoogle.com/assets/documents/gemini-2.5-pro.pdf>, 2025. Accessed: 2025-11-10. 4, 7
- [7] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The” something something” video database for learning and evaluating visual common sense. In *Proceedings of the IEEE international conference on computer vision*, pages 5842–5850, 2017. 2
- [8] Takeshi Kojima, Shixiang (Shane) Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems*, pages 22199–22213. Curran Associates, Inc., 2022. 5
- [9] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 1, 4
- [10] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, Limin Wang, and Yu Qiao. Mvbench: A comprehensive multi-modal video understanding benchmark, 2024. 1, 2
- [11] Yun Li, Yiming Zhang, Tao Lin, XiangRui Liu, Wenxiao Cai, Zheng Liu, and Bo Zhao. Sti-bench: Are mllms ready for precise spatial-temporal world understanding?, 2025. 2
- [12] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023. 4, 2, 3
- [13] Benlin Liu, Yuhao Dong, Yiqin Wang, Zixian Ma, Yansong Tang, Luming Tang, Yongming Rao, Wei-Chiu Ma, and Ranjay Krishna. Coarse correspondences boost spatial-temporal reasoning in multimodal language model. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 3783–3792, 2025. 5
- [14] Yunong Liu, Cristobal Eyzaguirre, Manling Li, Shubh Khanna, Juan Carlos Niebles, Vineeth Ravi, Saumitra Mishra, Weiye Liu, and Jiajun Wu. IKEA manuals at work: 4d grounding of assembly instructions on internet videos. In *NeurIPS Datasets and Benchmarks Track*, 2024. 2, 3, 8, 1
- [15] Zuyan Liu, Yuhao Dong, Ziwei Liu, Winston Hu, Jiwen Lu, and Yongming Rao. Oryx MLLM: On-Demand Spatial-Temporal Understanding at Arbitrary Resolution, 2025. *arXiv:2409.12961 [cs]*. 2
- [16] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. 1, 2
- [17] Shehan Munasinghe, Hanan Gani, Wenqi Zhu, Jiale Cao, Eric Xing, Fahad Shahbaz Khan, and Salman Khan. VideoGLaMM: A Large Multimodal Model for Pixel-Level Visual Grounding in Videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19036–19046, 2025. 2
- [18] OpenAI. Gpt-5 system card. <https://cdn.openai.com/gpt-5-system-card.pdf>, 2025. Accessed: 2025-10-19. 1, 2, 4, 3
- [19] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollar, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *The Thirteenth International Conference on Learning Representations*, 2025. 2, 7, 8, 5
- [20] Ye Sun, Hao Zhang, Henghui Ding, Tiehua Zhang, Xingjun Ma, and Yu-Gang Jiang. SAMA: Towards Multi-Turn Referential Grounded Video Chat with Large Language Models. *arXiv preprint arXiv:2505.18812*, 2025. 2
- [21] Dídac Surís, Sachit Menon, and Carl Vondrick. Vipergpt: Visual inference via python execution for reasoning. *Proceedings of IEEE International Conference on Computer Vision (ICCV)*, 2023. 2, 5, 7
- [22] Kexian Tang, Junyao Gao, Yanhong Zeng, Haodong Duan, Yanan Sun, Zhening Xing, Wenran Liu, Kaifeng Lyu, and Kai Chen. Lego-puzzles: How good are mllms at multi-step spatial reasoning? *arXiv preprint arXiv:2503.19990*, 2025. 2
- [23] Zongheng Tang, Yue Liao, Si Liu, Guanbin Li, Xiaojie Jin, Hongxu Jiang, Qian Yu, and Dong Xu. Human-centric spatio-temporal video grounding with visual transformers. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(12):8238–8249, 2021. 2
- [24] Gemini Team et al. Gemini: A family of highly capable multimodal models, 2025. 1, 4, 2

- [25] Huaxiaoyue Wang, Gonzalo Gonzalez-Pumariega, Yash Sharma, and Sanjiban Choudhury. Demo2code: From summarizing demonstrations to synthesizing code via extended chain-of-thought, 2023. 5
- [26] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023. 5, 6
- [27] Xingrui Wang, Wufei Ma, Angtian Wang, Shuo Chen, Adam Kortylewski, and Alan Yuille. Compositional 4d dynamic scenes understanding with physics priors for video question answering, 2025. 1, 2
- [28] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. In *Advances in Neural Information Processing Systems*, pages 28828–28857. Curran Associates, Inc., 2024. 2
- [29] Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. NExT-QA: Next Phase of Question-Answering to Explaining Temporal Actions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9777–9786, 2021. 1
- [30] Zihui Xue, Mi Luo, and Kristen Grauman. Seeing the arrow of time in large multimodal models. *arXiv preprint arXiv:2506.03340*, 2025. 2, 4, 5
- [31] Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-mark prompting unleashes extraordinary visual grounding in gpt-4v. *arXiv preprint arXiv:2310.11441*, 2023. 3
- [32] Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in Space: How Multimodal Large Language Models See, Remember and Recall Spaces. *arXiv preprint arXiv:2412.14171*, 2024. 2, 4, 6
- [33] Linli Yao, Haoning Wu, Kun Ouyang, Yuanxing Zhang, Caiming Xiong, Bei Chen, Xu Sun, and Junnan Li. Generative frame sampler for long video understanding. In *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025. Association for Computational Linguistics. 4
- [34] Yuqian Yuan, Hang Zhang, Wentong Li, Zesen Cheng, Boqiang Zhang, Long Li, Xin Li, Deli Zhao, Wenqiao Zhang, Yueteng Zhuang, Jianke Zhu, and Lidong Bing. VideoRefer Suite: Advancing Spatial-Temporal Object Understanding with Video LLM, 2025. *arXiv:2501.00599 [cs]*. 1, 2, 4, 5
- [35] Weichen Zhang, Ruiying Peng, Chen Gao, Jianjie Fang, Xin Zeng, Kaiyuan Li, Ziyong Wang, Jinqiang Cui, Xin Wang, Xinlei Chen, and Yong Li. The point, the vision and the text: Does point cloud boost spatial reasoning of large language models?, 2025. 2
- [36] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model, 2024. 1, 4
- [37] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data, 2024. 1, 4
- [38] Zhu Zhang, Zhou Zhao, Yang Zhao, Qi Wang, Huasheng Liu, and Lianli Gao. Where does it exist: Spatio-temporal video grounding for multi-form sentences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10668–10677, 2020. 2
- [39] Zhicheng Zheng, Xin Yan, Zhenfang Chen, Jingzhou Wang, Qin Zhi Eddie Lim, Joshua B. Tenenbaum, and Chuang Gan. Contphy: Continuum physical concept learning and reasoning from videos, 2024. 1, 2
- [40] Baichuan Zhou, Haote Yang, Dairong Chen, Junyan Ye, Tianyi Bai, Jinhua Yu, Songyang Zhang, Dahua Lin, Conghui He, and Weijia Li. Urbench: A comprehensive benchmark for evaluating large multimodal models in multi-view urban scenarios, 2025. 2
- [41] Honglu Zhou, Xiangyu Peng, Shrikant Kendre, Michael S. Ryoo, Silvio Savarese, Caiming Xiong, and Juan Carlos Niebles. Strefer: Empowering Video LLMs with Space-Time Referring and Reasoning via Synthetic Instruction Data, 2025. *arXiv:2509.03501 [cs]*. 1, 2
- [42] Shijie Zhou, Alexander Vilesov, Xuehai He, Ziyu Wan, Shuwang Zhang, Aditya Nagachandra, Di Chang, Dongdong Chen, Eric Xin Wang, and Achuta Kadambi. Vlm4d: Towards spatiotemporal awareness in vision language models. In *Proceedings of the IEEE/CVF international conference on computer vision*, 2025. 2
- [43] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, Hao Li, Jiahao Wang, Nianchen Deng, Songze Li, Yanan He, Tan Jiang, Jiapeng Luo, Yi Wang, Conghui He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Yingtong Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Han Lv, Lijun Wu, Kaipeng Zhang, Huipeng Deng, Jiaye Ge, Kai Chen, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhui Wang. Internv13: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. 1, 4, 5
- [44] Orr Zohar, Xiaohan Wang, Yann Dubois, Nikhil Mehta, Tong Xiao, Philippe Hansen-Estruch, Licheng Yu, Xiaofang Wang, Felix Juefei-Xu, Ning Zhang, Serena Yeung-Levy, and Xide Xia. Apollo: An Exploration of Video Understanding in Large Multimodal Models, 2024. *arXiv:2412.10360 [cs]*. 2